

PROJECT PROPOSAL

Medi-Assist

AI-Powered Online Pharmacy System

Software Engineering [C] | Spring 2025–2026

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1. Introduction

Medi-Assist is a web-based pharmaceutical management platform that is designed to change the way people take medicines and healthcare products in Bangladesh. Unlike standard e-commerce platforms, this system integrates AI-driven prescription analysis to make a bridge between medicines and people. This platform enables customers to search for medicines, describe their symptoms to an intelligent AI assistant, receive medicine recommendations, and place orders.

The system combines a rule-based Expert System with a Bayesian inference engine to generate preliminary medication recommendations based on user-reported symptoms. To ensure the highest standards of medical safety and accuracy, every AI-generated suggestion undergoes mandatory verification by a licensed pharmacist before confirmation. This approach not only mitigates prescription errors but also provides a reliable, scalable solution to the growing demand for convenient digital healthcare services in the region.

2. Problem Statement

The current pharmacy ecosystem in Bangladesh faces several critical challenges that Medi-Assist aims to solve:

Existing Problems

- **Inconvenient Access:** Even minor illnesses require patients to physically visit pharmacies, which causes undue hardship for the elderly and disabled.
- **Lack of Digital Guidance:** There isn't a reliable platform that uses AI to analyze symptoms and help patients before they talk to a pharmacist.
- **Inventory Inefficiency:** Customers must go to several places to get the medicine they need because the stock is inconsistent, and shortages happen often because there isn't a centralized system.

- **Unsafe Practices:** In rural areas, there aren't always enough qualified pharmacists, which leads to risky self-medication and mistakes that could have been avoided.
- **Fragmented Records:** Using paper prescriptions makes it more likely that records will be lost, forged, or given out wrong.

Medi-Assist solves these problems with a single digital platform that links patients, pharmacists, and delivery networks. The system makes sure that everyone can safely, efficiently, and easily manage their medications by combining an AI assistant for initial guidance with required professional verification.

3. Project Objectives

- The primary objectives of Medi-Assist are as follows:
- To provide customers with 24/7 access to medicines and AI-guided health assistance without requiring a physical pharmacy visit.
- To decrease medication distribution mistakes by allowing pharmacists to digitally check prescriptions and AI recommendations.
- To provide a structured delivery procedure that enables delivery staff to effectively manage and track allocated orders.
- To provide administrators with complete access and control over the platform, including system reports, inventories, and users.
- To implement an AI-based symptom analysis module that assists customers in identifying suitable medicines.
- To reduce medicine access barriers for people in areas with limited physical pharmacy availability.

4. Scope of the Project

The following functions and functionalities are covered by Medi-Assist for four different user roles:

4.1 Customer

- Register and login
- Browse medicine catalog by category, name, or ailment type.
- Use AI symptom checker, describe symptoms in text, and receive best medicine suggestions.
- Upload digital or scanned prescriptions for prescription only medicines.
- Add medicines to cart and place orders with delivery address.
- Track order status in real-time from confirmation to doorstep delivery.
- View order history and download invoices.
- Rate and review medicines and delivery experience.

User story: As a customer, I want to register and log in securely, search for medicines by name or category, describe my symptoms to the AI assistant and receive ranked medicine recommendations, place my order through

the cart, upload a prescription for prescription-only medicines, and track my delivery in real time. This allows me to conveniently access medicines and health guidance without visiting a pharmacy.

4.2 Pharmacist

- Login to the pharmacist dashboard using your validated login information.
- Examine AI-generated medication recommendations for every question from a consumer.
- Review AI suggestions.
- Check uploaded prescriptions for orders for prescription only medications.
- Manage the inventory of medications by adding, updating, and removing stock.
- After verifying prescriptions and AI suggestions, process and validate customer orders.
- Get notifications about low stock when medications are almost gone.

User story: As a pharmacist, I use a dashboard to review and verify AI-generated medicine suggestions for customers, approve or reject uploaded prescriptions, manage inventory, and confirm or cancel orders based on prescription validity. This ensures that all orders are medically safe, accurately fulfilled, and professionally verified before reaching the customer.

4.3 Delivery Person

- Login to view assigned delivery orders.
- Accept or deny delivery assignments depending on availability and location.
- Update the delivery status at each stage: picked up, transit, delivered.
- For each order, see the customer's delivery address and contact details.
- When orders are successfully handed over to the customer, mark them as delivered.

User story: As a delivery person, I need access to view assigned orders, customer addresses and contact details, update order status from pickup to delivery, and mark orders as successfully delivered. This allows for efficient management of deliveries and real-time tracking for customers.

4.4 Administrator

- Manage all user accounts, including consumers, pharmacists, and delivery of people.
- Track all orders, transactions, and AI suggestion records across the platform.
- Oversee the platform's medication library and medical categories.
- Create reports for sales, order volume, AI accuracy, and user activity.
- Set system settings, notification rules, and AI rule basis parameters.
- Respond to all forms of user complaints and support queries.

User story: As an administrator, my responsibilities include managing all user accounts, monitoring orders and AI recommendation logs, managing the medicine catalog and categories, generating sales and activity reports, and handling complaints and support requests to ensure the platform's security, efficiency, and reliability for all users.

5. Methodology

Medi-Assist will be built using the Agile Software Development Model, especially a Scrum-based iterative structure. This technique was chosen because the project's objectives, particularly those related to the AI recommendation engine, demand continuous testing, feedback, and iteration rather than a fixed sequential procedure.

Scrum Process

SCRUM comes from “**rugby**”, meaning teamwork brings the ball back into play. Scrum organizes the project into three phases:

1. **Pre-game:** Planning and setting up the project. A Product Backlog contains a list of all system requirements. Items are ranked according to their importance.
2. **Development (Game phase):** Actual work, sprint cycles, and product building. Work is done in Sprints (short development cycles). The system is developed in eight two-week sprints. Every sprint has a specific objective and a chosen list of backlog tasks to finish.
3. **Post-game:** Once the backlog items are completed, the system moves through final testing, review, and project closure.

Why Scrum Fits This Project

- The AI recommendation method requires testing and improvement after each iteration, which Scrum is suitable for.
- Requirements may change gradually as the system is created; Scrum adjusts through Sprint planning.
- Working features are supplied every two weeks rather than waiting for the finish.
- Risks have been identified and addressed early in each Sprint, rather than being found at the end.

6. System Requirements

6.1 Functional Requirements

- The system must allow users to register and log in using their email addresses and passwords.
- The system must allow role-based access for the following users: customer, pharmacist, delivery person, and administrator.
- Customers will be able to enter symptoms and obtain AI-based pharmaceutical recommendations.
- The AI module will provide suggestions based on an Expert System rule base and Bayesian scoring.
- The system must allow clients to upload prescription photos for verification.
- Pharmacists will be able to accept, amend, and reject AI ideas and prescriptions.
- The system must handle the whole order lifecycle: cart, checkout, confirmation, and delivery.
- Delivery personnel must update order status in real time from pickup to delivery.
- The administrator is responsible for managing users, drugs, categories, and reports.
- The system will notify users when the status of their orders changes.

6.2 Non-Functional Requirements

- **Performance:** Under typical load levels, the page load time should not exceed 3 seconds.
- **Security:** Encrypt all user data. Passwords are saved using bcrypt hashing.
- **Scalability:** The system must be capable of handling up to 1000 concurrent users without compromising performance.
- **Availability:** The system must be working at least 99% of the time.
- **Usability:** The interface must be responsive and functional on mobile, tablet, and desktop browsers.
- **Reliability:** AI recommendations must incorporate confidence scores and should not replace pharmacist judgment.
- **Maintainability:** The codebase must follow an MVC design and have specified API endpoints.

7. Technologies to be Used

The following technologies have been used for the development of Medi-Assist to ensure dependability, scalability, maintainability, and project fit.

Layer	Technology	Purpose
Frontend	React.js / HTML / CSS / Bootstrap	Responsive user interface
Backend	Node.js with Express.js	Server-side logic and REST API
Database	MySQL,	Relational data storage for users, medicines, orders
AI Module	Python	Custom rule-based expert system + Bayesian inference engine (probabilistic logic)
Authentication	JWT (JSON Web Tokens)	Secure session management
File Storage	Multer / Cloudinary	Prescription image upload and storage
Real-time Communication	Socket.IO	Live order status updates and notifications
Version Control	Git / GitHub	Source code management and collaboration
Deployment	Localhost	Application hosting

8. Project Timeline

Medi-Assist will be developed using the Agile (Scrum) approach, with eight two-week sprints spread over a total of 16 weeks.

Project Duration

Total: 16 weeks (approximately 4 months)

Sprint Duration: 2 weeks each

Number of phases: 8

Phase	Tasks	Duration	Week
Phase 1: Planning	Requirement gathering, system design, UML diagrams	2 weeks	Week 1–2
Phase 2: Setup	Database schema, authentication, role-based access	2 weeks	Week 3–4
Phase3: Core Features	Medicine catalog, customer dashboard, cart system	2 weeks	Week 5–6
Phase 4: AI Module	Expert system rules, Bayesian inference, symptom checker	2 weeks	Week 7–8
Phase5: Pharmacist Module	Verification workflow, prescription management	2 weeks	Week 9–10
Phase 6: Order & Delivery	Order pipeline, delivery tracking, notifications	2 weeks	Week 11–12
Phase 7: Admin & Reports	Admin dashboard, analytics, report generation	2 weeks	Week 13–14
Phase 8: Testing & Deploy	System testing, bug fixing, UI polish, deployment	2 weeks	Week 15–16

9. Budget Estimation

10. Conclusion

The Medi-Assist project, which developed an integrated AI-powered web pharmacy platform, successfully tackles major gaps in Bangladesh's pharmaceutical access and healthcare delivery environment. The system provides a practical, scalable, and patient-centered solution to long-standing challenges such as inconvenient physical access, unsafe self-medication, prescription errors, and fragmented medicine availability by strategically combining artificial intelligence for preliminary symptom analysis, mandatory pharmacist verification for safety, and efficient digital workflows for ordering, inventory management, and delivery.

The Agile (Scrum) approach enabled iterative development, continuous improvement of the AI recommendation engine, and timely delivery of functional modules within the needed timeframe. The solution uses a current, JavaScript-centric technological stack in conjunction with a relational database and real-time communication protocols to provide responsiveness, security, and maintainability. The iterative development methodology allowed for continual refining of the recommendation engine and user processes, culminating in a workable prototype that combines automation with expert oversight.

The system achieves its key design goals with a single web interface that enables end-to-end drug management. Customers benefit from 24/7 availability, simple symptom description input, graded AI-generated recommendations, digital prescription filing, faster order fulfillment, and real-time delivery monitoring. Pharmacists have specific tools for reviewing AI recommendations, validating prescriptions, and managing inventory via proactive low-stock warnings. Delivery workers benefit from rapid job assignment and status update capabilities, while administrators have full control over users, transactions, system performance, and AI recommendation logs. These characteristics, taken together, address inventory fragmentation, prescription forgery threats, and distribution of inefficiencies that exist in today's economy.

In short, Medi-Assist is a realistic step toward more inclusive digital healthcare services. The technology helps to make pharmaceutical delivery safer, more efficient, and egalitarian by integrating artificial intelligence for accessibility with pharmacist experience for dependability. The entire system serves as a fundamental model capable of enabling future upgrades and expanded implementation in resource-constrained healthcare situations.